7 Hypotheticals and You Testing Your Questions: Difficulty Scale ☺ ☺ ☺½ (don't plan on going out tonight)

What you'll learn about in this chapter

- ✦ Understanding the difference between a sample and a population
- ✦ Understanding the importance of the null and research hypotheses
- ✦ Using criteria to judge a good hypothesis

So You Want to Be a Scientist . . .

You might have heard the term *hypothesis* used in other classes. You may even have had to formulate one for a research project you did for another class, or you may have read one or two in a journal article. If so, then you probably have a good idea what a hypothesis is. For those of you who are unfamiliar with this often-used term, a **hypothesis** is basically "an educated guess." Its most important role is to reflect the general problem statement or question that was the motivation for asking the research question in the first place.

Introduction to Chapter 7

That's why taking the care and time to formulate a really precise and clear research question is so important. This research question will guide your creation of a hypothesis, and in turn, the hypothesis will determine the techniques you will use to test it and answer the question that was originally asked.

So, a good hypothesis translates a problem statement or a research question into a form that is more amenable to testing. This form is called a hypothesis. We will talk about what makes a hypothesis a good one later in this chapter. Before that, let's turn our attention to the difference between a sample and a population. This is an important distinction, because hypothesis testing deals with a sample and then the results are generalized to the larger population. We also address the two main types of hypotheses (the null hypothesis and the research hypothesis). But first, let's formally define some simple terms that we have used

earlier in *Statistics for People Who (Think They) Hate Statistics*.

Samples and Populations

As a good scientist, you would like to be able to say that if Method A is better than Method B in your study, this is true forever and always and for all people in the universe, right? Indeed. And, if you do enough research on the relative merits of Methods A and B and test enough people, you may someday be able to say that.

Looking at Sampling Error

But don't get too excited, because it's unlikely you will ever be able to speak with such confidence. It takes too much money ($$$) and too much time (all those people!) to do all that research, and besides, it's not even necessary. Instead, you can just select a representative sample from the population and test your hypothesis about the relative merits of Methods A and B.

Given the constraints of never enough time and never enough research funds, with which almost all scientists live, the next best strategy is to take a portion of a larger group of participants and do the research with that smaller group. In this context, the larger group is referred to as a *population*, and the smaller group selected from that population is referred to as a *sample*.

A measure of how well a sample approximates the characteristics of a population is called **sampling error**. Sampling error is basically the difference between the values of the sample statistic and the population parameter. The higher the sampling error, the less precision you have in sampling, and the more difficult it will be to make the case that what you find in the sample indeed reflects what you expected to find in the population. And just as there are measures of variability regarding distributions, so there are measures of the variability of this difference between a sample measure and a population measure. This is often called the *standard error*—it's basically the standard deviation of the difference between those two measures.

Samples should be selected from populations in such a way that the sample matches as closely as possible the characteristics of the population. The goal is to have the sample be as much like the population as possible. The most important implication of ensuring similarity between the two is that the research results based on the sample can be

generalized to the population. When the sample accurately represents the population, the results of the study are said to have a high degree of generalizability.

A high degree of generalizability is an important quality of good research because it means that the time and effort (and $$$) that went into the research may have implications for groups of people other than the original participants.

It's easy to equate "big" with "representative." Keep in mind that it is far more important to have an accurately representative sample than it is to have a big sample (people often think that big is better—only true on Thanksgiving, by the way). Having lots and lots of participants in a sample may be very impressive, but if the participants do not represent the larger population, the research will have little value.

The Null Hypothesis

Okay. So we have a sample of participants selected from a population, and to begin the test of our research hypothesis, we first formulate the **null hypothesis**.

The Null Hypothesis

The null hypothesis is an interesting little creature. If it could talk, it would say something like "I represent no relationship between the variables that you are studying." In other words, null hypotheses are statements of equality demonstrated by the following real-life null hypotheses taken from a variety of popular social and behavioral science journals. Names have been changed to protect the innocent.

- There will be *no difference* between the average score of 9th graders and the average score of 12th graders on the ABC memory test.
- There is *no difference* between the effectiveness of community-based, long-term care and the effectiveness of in-home, long-term care in promoting the social activity of older adults when measured using the Margolis Scale of Social Activities.
- There is *no relationship* between reaction time and problem-solving ability.
- There is *no difference* between white and black families in the amount of assistance families offer their children in school-related activities.

What these four null hypotheses have in common is that they all contain a statement that two or more things are equal or unrelated (that's the "no difference" and "no relationship" part) to each other.

The Purposes of the Null Hypothesis

What are the basic purposes of the null hypothesis? The null hypothesis acts as both a starting point and a benchmark against which the actual outcomes of a study can be measured.

Let's examine each of these purposes in more detail.

First, the null hypothesis acts as a starting point because it is the state of affairs that is accepted as true in the absence of any other information. For example, let's look at the first null hypothesis we stated earlier:

> There will be no difference between the average score of 9th graders and the average score of 12th graders on the ABC memory test.

Given absolutely no other knowledge of 9th and 12th graders' memory skills, you have no reason to believe that there will be differences between the two groups, right? If you know nothing about the relationship between these variables, the best you can do is guess. And that's taking a chance. You might speculate as to why one group might outperform another, but if you have no evidence a priori (before the fact), then what choice do you have but to assume that they are equal?

This lack of a relationship as a starting point is a hallmark of this whole topic. In other words, until you prove that there is a difference, you have to assume that there is no difference. And a statement of no difference or no relationship is exactly what the null hypothesis is all about. Such a statement assures us (as members of the scientific community) that we are starting on a level playing field with no bias toward one or the other direction as to how the test of our hypothesis will turn out.

Furthermore, if there are any differences between these two groups, you have to assume that these differences are due to the most attractive explanation for differences between any groups on any variable—chance! That's right: Given no other information, chance is always the most likely and attractive explanation for the observed differences between two groups or the relationship between variables. Chance explains what we cannot. You might have thought of chance as the odds of winning that $5,000 nickel jackpot at the slots, but we're talking about chance as all that other "stuff" that clouds the picture and makes it even more difficult to understand the "true" nature of relationships between variables.

For example, you could take a group of soccer players and a group of football players and compare their running speeds, thinking about whether playing soccer or playing football

makes athletes faster. But look at all the factors we don't know about that could contribute to differences. Who is to know whether some soccer players practice more, or whether some football players are stronger, or whether both groups are receiving additional training of different types?

What's more, perhaps the way their speed is being measured leaves room for chance; a faulty stopwatch or a windy day can contribute to differences unrelated to *true* running speed. As good researchers, our job is to eliminate chance factors from explaining observed differences and to evaluate other factors that might contribute to group differences, such as intentional training or nutrition programs, and see how they affect speed.

The point is, if we find differences between groups and the differences are not due to training, we have no choice but to attribute the difference to chance. And, by the way, you might find it useful to think of *chance* as being somewhat equivalent to the idea of *error*. When we can control sources of error, the likelihood that we can offer a meaningful explanation for some outcome increases.

The second purpose of the null hypothesis is to provide a benchmark against which observed outcomes can be compared to see if these differences are due to some other factor. The null hypothesis helps to define a range within which any observed differences between groups can be attributed to chance (which is the null hypothesis's contention) or are due to something other than chance (which perhaps would be the result of the manipulation of some variable, such as training in our example of the soccer and football players).

Most research studies have an implied null hypothesis, and you may not find it clearly stated in a research report or journal article. Instead, you'll find the research hypothesis clearly stated, and this is where we now turn our attention.

The Research Hypothesis

Whereas a null hypothesis is a statement of no relationship between variables, a **research hypothesis** is a definite statement that a relationship exists between variables. For example, for each of the null hypotheses stated earlier, here is a corresponding research hypothesis. Notice that we said "a" and not "the" corresponding research hypothesis because there certainly could be more than one research hypothesis for any one null hypothesis.

The Research Hypothesis

- The average score of 9th graders *is different* from the average score of 12th graders on the ABC memory test.
- The effectiveness of community-based, long-term care *is different* from the effectiveness of in-home, long-term care in promoting the social activity of older adults when measured using the Margolis Scale of Social Activities.
- Slower reaction time and problem-solving ability *are positively related.*
- There *is a difference* between white and black families in the amount of assistance families offer to their children in school-related activities.

Each of these four research hypotheses has one thing in common: They are all statements of inequality. They posit a relationship between variables and not an equality, as does the null hypothesis.

The nature of this inequality can take two different forms—a directional or a nondirectional research hypothesis. If the research hypothesis posits no direction to the inequality (such as "different from"), the hypothesis is a nondirectional research hypothesis. If the research hypothesis posits a direction to the inequality (such as "more than" or "less than"), the research hypothesis is a directional research hypothesis.

The Nondirectional Research Hypothesis

A **nondirectional research hypothesis** reflects a difference between groups, but the direction of the difference is not specified.

For example, the research hypothesis . . .

> The average score of 9th graders is different from the average score of 12th graders on the ABC memory test

. . . is nondirectional in that the direction of the difference between the two groups is not specified. The hypothesis is a research hypothesis because it states that there is a difference, and it is nondirectional because it says nothing about the direction of that difference.

A nondirectional research hypothesis, like this one, would be represented by the following equation:

(7.1)
H 1 : X ¯ 9 ≠ X ¯ 12 ,

$$H_1: \overline{X}_9 \neq \overline{X}_{12}, \qquad (7.1)$$

where

- H_1 represents the symbol for the first (of possibly several) research hypotheses;
- X ¯ 9 $\overline{X}_9$ represents the average memory score for the sample of 9th graders;
- X ¯ 12 $\overline{X}_{12}$ represents the average memory score for the sample of 12th graders; and
- ≠ means "is not equal to."

The Directional Research Hypothesis

A **directional research hypothesis** reflects a difference between groups, and the direction of the difference is specified.

For example, the research hypothesis . . .

> The average score of 12th graders is greater than the average score of 9th graders on the ABC memory test

. . . is directional because the direction of the difference between the two groups is specified. One is hypothesized to be greater than (not just different from) the other.

Examples of two other directional hypotheses are these:

> *A* is greater than *B* (or *A* > *B*).
> *B* is greater than *A* (or *A* < *B*).

These both represent inequalities (greater than or less than). A directional research hypothesis such as the one described above, where 12th graders are hypothesized to score better than 9th graders, would be represented by the following equation:

(7.2)
H 1 : X ¯ 12 > X ¯ 9 ,

$$H_1: \overline{X}_{12} > \overline{X}_9, \qquad (7.2)$$

where

- H_1 represents the symbol for the first (of possibly several) research hypotheses;
- X ¯ 9 $\overline{X}_9$ represents the average memory score for the sample of 9th graders;
- X ¯ 12 $\overline{X}_{12}$ represents the average memory score for the sample of 12th graders; and
- > means "is greater than."

What is the purpose of the research hypothesis? It is this hypothesis that is directly tested as an important step in the research process. The results of this test are compared with what you expect by chance alone (reflecting the null hypothesis) to see which of the two is the more attractive explanation for any differences between groups or variables you might observe.

Table 7.1 gives the four null hypotheses and accompanying directional and nondirectional research hypotheses.

Another way to talk about directional and nondirectional hypotheses is to talk about one- and two-tailed tests. A **one-tailed test** (reflecting a directional hypothesis) posits a difference in a particular direction, such as when we hypothesize that Group 1 will score higher than Group 2. A **two-tailed test** (reflecting a nondirectional hypothesis) posits a difference but in no particular direction. The importance of this distinction begins when you test different types of hypotheses (one- and two-tailed) and establish probability levels for rejecting or not rejecting the null hypothesis. More about this in Chapter 9. Promise.

Table 7.1 Null Hypotheses and Corresponding Research Hypotheses

Null Hypothesis	Nondirectional Research Hypothesis	Directional Research Hypothesis
There will be **no difference** in the average score of 9th graders and the average score of 12th graders on the ABC memory test.	Twelfth graders and 9th graders will **differ** on the ABC memory test.	Twelfth graders will have a **higher** average score on the ABC memory test than will 9th graders.
There is **no difference** between the effectiveness of community-based, long-term care for older adults and the effectiveness of in-home, long-term care for older adults when measured using the Margolis Scale of Social Activities.	The effect of community-based, long-term care for older adults is **different** from the effect of in-home, long-term care for older adults when measured using the Margolis Scale of Social Activities.	Older adults exposed to community-based, long-term care score **higher** on the Margolis Scale of Social Activities than do older adults receiving in-home, long-term care.
There is **no relationship** between reaction time and problem-solving ability.	There **is a relationship** between reaction time and problem-solving ability.	There **is a positive relationship** between reaction time and problem-solving ability.
There is **no difference** between white and black families in the amount of assistance families offer their children in educational activities.	The amount of assistance offered by white families to their children in educational activities is **different** from the amount of support offered by black families to their children in educational activities.	The amount of assistance offered by white families to their children in educational activities is **more** than the amount of support offered by black families to their children in educational activities.

Some Differences Between the Null Hypothesis and the Research Hypothesis

Besides the null hypothesis representing an equality and the research hypothesis representing an inequality, the two types of hypotheses differ in several other important ways.

Questions About Test Direction

First, for a bit of review, the two types of hypotheses differ in that one (the null hypothesis) states that there is no relationship between variables (an equality), whereas the research hypothesis states that there is a relationship between the variables (an inequality). This is the primary difference.

Second, null hypotheses always refer to the population, whereas research hypotheses always refer to the sample. We select a sample of participants from a much larger population. We then try to generalize the results from the sample back to the population. If you remember your basic philosophy and logic (you did take these courses, right?), you'll remember that going from small (as in a sample) to large (as in a population) is a process of inference.

Third, because the entire population cannot be directly tested (again, it is impractical, uneconomical, and often impossible), you can't say with 100% certainty that there is no real difference between segments of the population on some variable. Rather, you have to infer it (indirectly) from the results of the test of the research hypothesis, which is based on the sample. Hence, the null hypothesis must be indirectly tested, and the research hypothesis can be directly tested.

Fourth, the null hypothesis is written with an equal sign, while the research hypothesis is written with a not equal to, greater than, or less than sign.

Fifth, null hypotheses are always written using Greek symbols, and research hypotheses are always written using Roman symbols. Thus, the null hypothesis that the average score for 9th graders is equal to that of 12th graders is represented like this:

(7.3)
H 0 : μ 9 = μ 12 ,

$$H_0: \mu_9 = \mu_{12}, \qquad (7.3)$$

where

- H_0 represents the null hypothesis;
- μ_9 represents the theoretical average for the population of 9th graders; and
- μ_{12} represents the theoretical average for the population of 12th graders.

The research hypothesis that the average score for a sample of 12th graders is greater than the average score for a sample of 9th graders is shown in Formula 7.2 (on page 139).

Finally, because you cannot directly test the null hypothesis, it is an *implied* hypothesis. But the research hypothesis is explicit and is stated as such. This is another reason why you rarely see null hypotheses stated in research reports and will almost always see a statement (be it in symbols or words) of the research hypothesis.

What Makes a Good Hypothesis?

You now know that hypotheses are educated guesses—a starting point for a lot more to come. As with any guess, some hypotheses are better than others right from the start. We can't stress enough how important it is to ask the question you want answered and to keep in mind that any hypothesis you present is a direct extension of the original question you asked. This question will reflect your personal interests and motivation and your understanding of what research has been done previously. With that in mind, here are criteria you might use to decide whether a hypothesis you read in a research report or one that you formulate is acceptable.

To illustrate, let's use an example of a study that examines the effects of providing afterschool child care to employees who work late on the parents' adjustment to work. Here is a well-written directional research hypothesis:

> Parents who enroll their children in afterschool programs will miss fewer days of work in 1 year and will have a more positive attitude toward work, as measured by the Attitude Toward Work survey, than will parents who do not enroll their children in such programs.

Here are the criteria.

First, a good hypothesis is stated in declarative form and not as a question. While the above hypothesis may have started in the researcher's mind as the question "Are parents better employees . . . ?" it was not posed because hypotheses are most effective when they make a clear and forceful statement.

Second, a good hypothesis posits an expected relationship between variables. The hypothesis in our example clearly describes the expected relationships among afterschool child care, parents' attitude, and absentee rate. These variables are being tested to see if one (enrollment in the afterschool program) has an effect upon the others (absentee rate and attitude).

Notice the word *expected* in the above criterion. Defining an expected relationship is intended to prevent a fishing trip to look for any relationships that may be found

(sometimes called the "shotgun" approach), which may be tempting but is not very productive. You do get somewhere using the shotgun approach, but because you don't know where you started, you have no idea where you end up.

In the fishing-trip approach, you throw out your line and take anything that bites. You collect data on as many things as you can, regardless of your interest or even whether collecting the data is a reasonable part of a scientific investigation. Or, to use a shotgun analogy, you load up them guns and blast away at anything that moves, and you're bound to hit something. The problem is, you may not want what you hit, and worse, you may miss what you want to hit, and worst of all (if possible), you may not know what you hit! Big data and data mining (see Chapter 17) anyone? Good researchers do not want just anything they can catch or shoot. They want specific results. To get them, researchers need their opening questions and hypotheses to be clear, forceful, and easily understood.

Third, hypotheses reflect the theory or literature on which they are based. As you read in Chapter 1, the accomplishments of scientists rarely can be attributed to just their own hard work. Their accomplishments are always due, in part, to many other researchers who came before them and laid the framework for later explorations. A good hypothesis reflects this, in that it has a substantive link to existing literature and theory. In the aforementioned example, let's assume that there is literature indicating that parents are more comfortable knowing their children are being cared for in a structured environment and parents can then be more productive at work. Knowing this would allow one to hypothesize that an afterschool program would provide the security parents are looking for. In turn, this allows them to concentrate on working rather than calling or texting to find out whether Rachel or Gregory got home safely.

Fourth, a hypothesis should be brief and to the point. You want your hypothesis to describe the relationship between variables in a declarative form and to be as direct and explicit as possible. The more to the point it is, the easier it will be for others (such as your master's thesis or doctoral dissertation committee members!) to read your research and understand exactly what you are hypothesizing and what the important variables are. In fact, when people read and evaluate research (as you will learn more about later in this chapter), the first thing many of them do is find the hypotheses to get a good idea as to the general purpose of the research and how things will be done. A good hypothesis tells you both of these things.

Fifth, good hypotheses are testable hypotheses—and testable hypotheses contain variables that can be measured. This means that you can actually carry out the intent of the question reflected by the hypothesis. You can see from our example hypothesis that the important

comparison is between parents who have enrolled their child in an afterschool program and those who have not. Then, such things as attitude and workdays missed will be measured. Identifying these two groups of parents and measuring the variables attitude and absenteeism are both reasonable objectives. Attitude is measured by the Attitude Toward Work survey (a fictitious title, but you get the idea), and absenteeism (the number of days missed at work) is an easily recorded and unambiguous measure almost always available in payroll records (assuming you've been granted access). Think how much harder things would be if the hypothesis were stated as *Parents who enroll their children in afterschool care feel better about their jobs.* Although you might get the same message, the results might be more difficult to interpret given the ambiguous nature of the phrase *feel better*.

In sum, hypotheses should . . .

Creating Hypotheses

- be stated in declarative form,
- posit a relationship between variables,
- reflect a theory or a body of literature on which they are based,
- be brief and to the point, and
- be testable.

When a hypothesis meets each of these five criteria, you know that it is good enough to continue with a study that will accurately test the general question from which the hypothesis was derived.

Real-World Stats

You might think that the scientific method and the use of null and research hypotheses is beyond question in the world of scientific research. Well, you would be wrong. Here's just a sampling of some of the articles published in professional journals over the past few years that raise concerns. No one is yet ready to throw out the scientific method as the best approach for testing hypotheses, but it's not a bad idea to every now and then question whether the method always and forever is the best model to use.

Want to know more? Go online or to the library and find . . .

• Jeff Gill from California Polytechnic State University raises a variety of issues that call into question the use of the null hypothesis significance-testing model as the best way to

evaluate hypotheses. He focuses on political science (his area) and how the use of the technique is widely misunderstood. Major problems are discussed and some solutions offered. You can find the article by looking for this reference: Gill, J. (1999). The insignificance of null hypothesis testing. *Politics Research Quarterly, 52*, 647–674.

Real-World Stats

• Howard Wainer and Daniel Robinson take these criticisms one step further and suggest that the historical use of such procedures was reasonable but that modifications to significance testing and the interpretations of outcomes would serve modern science well. Basically, they are saying that other tools (such as effect size, which we discuss in Chapter 11) should be used to evaluate outcomes. Read all about it in Wainer, H., & Robinson, D. H. (2003). Shaping up the practice of null hypothesis significance testing. *Educational Researcher, 32*, 22–30.

More Real-World Stats

• Finally, in the really interesting article "A Vast Graveyard of Undead Theories: Publication Bias and Psychological Science's Aversion to the Null," Christopher Ferguson and Moritz Heene raise the very real issue that many journals refuse to publish outcomes where a null result is found (such as no difference between groups). They believe that when such outcomes are not published, false theories are never taken to task and never tested for their truthfulness. Thus, the replicability of science (a very important aspect of the entire scientific process) is compromised. You can find more about this in Ferguson, C. J., & Heene, M. (2012). A vast graveyard of undead theories: Publication bias and psychological science's aversion to the null. *Perspectives on Psychological Science, 7*, 555–561.

Even More Real-World Stats

We hope you get the idea from the above examples that science is not black-and-white, cut-and-dried, or any other metaphor indicating there is only a right way and only a wrong way of doing things. Science is an organic and dynamic process that is always changing in its focus, methods, and potential outcomes.

Summary

Chapter Summary

A central component of any scientific study is the hypothesis, and the different types of hypotheses (null and research) help form a plan for answering the questions asked by the purpose of our research. The null hypothesis provides a starting point and benchmark for research, and we use it as a comparison as we evaluate the acceptability of the research hypothesis. Now let's move on to how null hypotheses are actually tested.

Time to Practice

1. Go to the library and select five empirical research articles (those that contain actual data) from your area of interest. For each one, list the following:
    1. What is the null hypothesis (implied or explicitly stated)?
    2. What is the research hypothesis (implied or explicitly stated)?
    3. And what about those articles with no hypothesis clearly stated or implied? Identify those articles and see if you can write a research hypothesis for them.
2. While you're at the library, select two other articles from an area in which you are interested and write a brief description of the sample and how it was selected from the population. Be sure to include some words about whether the researchers did an adequate job of selecting the sample; be able to justify your answer.
3. For the following research questions, create one null hypothesis, one directional research hypothesis, and one nondirectional research hypothesis.
    1. What are the effects of attention span on out-of-seat classroom behavior?
    2. What is the relationship between the quality of a marriage and the quality of the spouses' relationships with their siblings?
    3. What is the best way to treat an eating disorder?
4. Go back to the five hypotheses that you found in question 1 above and evaluate each using the five criteria that were discussed at the end of the chapter.
5. What kinds of problems might using a poorly written or ambiguous research hypothesis introduce?
6. What is the null hypothesis, and what is one of its important purposes? How does it differ from the research hypothesis?
7. What is *chance* in the context of a research hypothesis? And what do we do about chance in our experiments?
8. Why does the null hypothesis presume no relationship between variables?

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